

Assignment-midterm-assignment

Midterm Assignment: AI System Analysis

Overview

Identify a real-world AI system (currently in use) and write a comprehensive analysis examining its societal impact, ethical implications, and policy considerations.

Assignment Details

Due Date: Week 7

Length: 1,500-2,000 words

Format: Word document or PDF with citations

Submission: D2L Brightspace

Instructions

Step 1: Select an AI System

Choose one of the following or propose your own: - Facial recognition systems (law enforcement, retail) - Hiring/recruitment algorithms - Social media recommendation algorithms - Credit scoring and lending algorithms - Healthcare diagnosis systems - Autonomous vehicles - Content moderation systems (Meta, YouTube, etc.)

Step 2: Research the System

Gather information about: - How the system works (technical overview) - Where it's deployed and at scale - Who created it and maintains it - Intended purpose vs. actual use

Step 3: Analyze Five Key Dimensions

Write sections addressing each:

A. Functionality & Limitations

- What does this AI system do?
- What are its stated capabilities and limitations?
- When does it fail or make mistakes?

B. Bias & Fairness

- What historical or systematic biases might affect this system?
- Who benefits from this system? Who might be harmed?
- Are there documented cases of discriminatory outcomes?

C. Privacy & Data Ethics

- What data does the system collect?
- How is data stored, used, and shared?
- What are the privacy implications?

D. Social & Economic Impact

- How does this system affect individuals and communities?
- What are employment or economic implications?
- Are there societal winners and losers?

E. Policy & Governance

- What regulations (if any) govern this system?
- What policy gaps exist?
- What recommendations would you make?

Rubric

Criteria	Excellent	Good	Satisfactory	Needs Work
System Understanding	Clear, accurate technical explanation	Good grasp of how system works	Basic explanation provided	Incomplete or inaccurate
Research Quality	Multiple credible sources; well-integrated	Good sources with solid integration	Adequate sources with limited depth	Few sources or poor quality

Criteria	Excellent	Good	Satisfactory	Needs Work
Critical Analysis	Sophisticated examination of all dimensions	Thoughtful analysis of most areas	Covers main points; surface level	Superficial or incomplete
Ethical Reasoning	Nuanced exploration of tradeoffs and implications	Identifies ethical issues; considers multiple perspectives	Ethical issues noted	Misses key ethical concerns
Organization & Writing	Clear structure; excellent writing	Well-organized; good writing	Understandable; minor issues	Disorganized or unclear
Citations	Proper APA citations throughout	Generally correct citations	Some citations provided	Missing or improper citations

Resources

- algorithmic.org - Algorithmic Accountability Resource
- ProPublica Data Stories - Investigative journalism on AI
- News outlets: MIT Technology Review, Wired, The Verge
- Academic journals: arXiv, AI & Society, AI Ethics

Tips for Success

1. **Choose a system you're curious about** - Research is easier when genuinely interested
2. **Start with news coverage** - Get overview before diving into technical details
3. **Look for controversies** - Published debates reveal ethical dimensions
4. **Use real examples** - Cite specific incidents or cases
5. **Show multiple perspectives** - Don't oversimplify; acknowledge complexity

Accessibility Accommodations

- Request extended deadline if needed
- Can submit as audio essay or video essay (contact instructor)
- Can work in pairs if preferred
- Instructor can provide source summaries

Questions? Post in the discussion forum or email your instructor.